



Review article

Effects of exercise interventions on cognitive functions in healthy populations: A systematic review and meta-analysis

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ABSTRACT

Chronic exercise intervention is a non-pharmacological therapy suggested to improve cognitive function in various populations. However, few meta-analyses have assessed the cognitive benefits associated with all FITT-VP variables (exercise frequency, intensity, bout duration time, type, volume or total intervention length, and progression) in healthy populations by age. Thus, this meta-analysis assessed the effects of each FITT-VP variable on cognitive function in healthy children, adults, and older adults. This study followed PRISMA guidelines. After searching PubMed and the Web of Science, we included 54 randomized controlled trials to examine the effects of FITT-VP variables on five cognitive domains: global cognition, executive function, memory, attention, and information processing. Moderation analyses assessed the effects by age and by each exercise variable. Exercise benefitted overall cognition and all subcognitive domains. Aerobic and resistance exercise showed the greatest benefits on global cognition and executive function respectively, whereas mind-body exercise benefitted memory. Among all populations, older adults showed the greatest benefits of exercise on global cognition, executive function, and memory compared with controls. Additional studies are needed to assess the effects of exercise on attention and information processing. This meta-analysis offers new insights on the relationships between cognition and FITT-VP exercise variables in healthy populations.

1. Introduction

The problem of the cognitive decline that accompanies aging is becoming more serious with the world's aging population, and how to deal with this problem more economically and effectively is getting more and more attention. Exercise is increasingly being recognized as a promising non-pharmacological method to maintain, improve, and even prevent the decline of human cognitive functions (Huang et al., 2022).

According to the Physical Activity Guidelines for Americans, widespread cognitive performance benefits of regular physical activity (PA) have been described for diverse populations (Etnier and Chang, 2019). Meanwhile, a hand of review studies also have been offered the multiple evidence to support the cognitive benefits from participating in chronic PA or exercise among children (Li et al., 2020), adolescents and young adults (Bento-Torres et al., 2019), and older adults (Kelly et al., 2014; Wayne et al., 2014; Chang et al., 2012). Among these populations in different life stages, a great deal of review interests have been directed to the effects of exercise interventions on cognition in specific populations. For example, the older population with Alzheimer's disease or mild

cognitive impairment (Huang et al., 2022; Chen et al., 2020a; Brini et al., 2018), children or adolescences individuals with attention-deficit/hyperactivity disorder (Catalá-López et al., 2017; Seiffer et al., 2022; Zhang et al., 2020), and adults persons with substance abuse (Morais et al., 2018; Ashdown-Franks et al., 2020; AshaRani et al., 2020), correspondingly, all of these specific populations could obtain varying degrees of cognitive benefits from participating in diverse chronic PA or exercise. However, these evidence of cognitive benefits from the specific populations have been too narrowly focused to be applicable to the general population. Although there was a recent review study of healthy populations found that exercise improved cognition generally rather than domain-specific (Ludyga et al., 2020), the cognitive performance benefit difference among different age stages were less known through their results and conclusions. Furthermore, the other obvious problem we found based on above studies is that most studies are much more focused on a particular aspect of cognition, a single age group, and a type of chronic exercise training rather than an overall assessment. Considering these issues, it is necessary to conduct a comprehensive study to understand the various cognitive performance

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benefits from chronic exercise in healthy individuals across their lifespan.

In addition, experimental and methodological shortcomings are apparent in previous meta-analyses. As we mentioned above, those studies typically compared the effects of one exercise characteristic, such as the type of exercise type (e.g., aerobic exercise, resistance exercise, or tai chi) on cognitive performance. For example, one previous meta-analysis examined the impact of aerobic exercise, resistance training, and tai chi on cognitive function among older adults who had no known cognitive impairment, but no exercise frequency or intensity effects were detected (Kelly et al., 2014). However, guidelines from the American College of Sports Medicine (ACSM) indicate that several exercise variables should be assessed when considering exercise prescriptions to improve the cognitive health of the brain; they proposed the FITT-VP principle as a reference, which is defined as exercise frequency (how often), intensity (difficulty), time (duration of each bout of exercise), type (of exercise), volume (total amount of exercise per intervention), and progression (change in difficulty in an exercise program over intervention time; for example, an intervention that begins at moderate intensity, and then progresses to high intensity over a number of weeks) (Chen et al., 2020a; Piercy et al., 2018; Piercy and Troiano, 2018). Given the importance of understanding the effects that each of these exercise variables may exert on cognitive functions, a systematic review and meta-analysis that includes the FITT-VP variables is needed to advance this understanding.

In summary, there is dearth of studies that have simultaneously considered: (1) whether chronic exercise interventions may affect various cognitive functions of individuals in the general population from childhood to adulthood and into older age, and (2) how each of exercise variables further moderating this relationship. Therefore, the present systematic review and meta-analysis aimed to examine the effects of chronic exercise interventions on cognitive function in healthy populations of children and youths (ages 6–17 years old), adults (ages 18–60 years old), and elderly adults (ages >60 years) by assessing studies that used a RCT design. We assessed the effects of exercise overall on numerous cognitive domains and conducted moderation analyses to examine the contributions of FITT-VP exercise variables and age. While there are elements (cognitive domains, different age groups or exercise intervention principles) of this review explored separately in other reviews, this review will still serve a purpose as a comprehensive "one stop shop" and offer new insights into how chronic exercise interventions affect cognitive functions during the process of aging in the general population.

2. Methods

We pre-registered our study at PROSPERO (CRD42022383441). This systematic review and meta-analysis followed the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guideline (Page et al., 2021) and was conducted in accordance with the Cochrane Handbook for Systematic Reviews of Interventions (Higgins, 2022). The PRISMA checklist presented in **Supplement 1**. This study was approved by the Ethics Committee of Shanghai University of Sport.

2.1. Search strategy

The systematic search was conducted using two databases (Web of Science and PubMed) from January, 2003 through April, 2022. We first developed the search strategy for Medical Subject Heading (MeSH) terms and entry terms through PubMed, and the strategy was reviewed by experts from the fields of exercise and cognition for its comprehensiveness and accuracy. In addition, we conducted a "vertical" search of the references listed in the retrieved meta-analyses. The details of this search strategy are presented in **Supplement 2**.

2.2. Inclusion/exclusion criteria

All the included studies met the following inclusion criteria: (1) The study was an RCT with healthy human participants conducted to explore the effects of chronic exercise interventions on cognitive function and was reported in English. (2) The interventions could include any type of exercise training with planned supervision. (3) Participants in the control group received no intervention, usual care, health education, sham exercise training, or were on a wait list for the study. (5) Studies had to report at least one cognitive outcome that included global cognition, executive function, memory, attention, or information processing. Studies were excluded based on the following criteria: (1) The study involved exercise training interventions combined with other non-exercise activities, such as cognitive training or video games. (2) The study focus was on acute exercise interventions. (3) Participants had a cognitive impairment or other physiological or psychiatric illness, such as Parkinson disease, attention-deficit/hyperactivity disorder, epilepsy, multiple sclerosis, diabetes, autism, or schizophrenia. (4) Studies are not in English and the full-text unavailable.

After the removal of duplicate studies, study selection was independently screened by two authors (JF-J and Y-Y), including the titles, abstracts, and keywords that were relevant to the eligible studies. If a disagreement could not be resolved after discussion between the two authors, a third author (LP-Z) joined the discussion to reach a consensus.

2.3. Data extraction and coding

We extracted data on study characteristics, exercise prescriptions, and cognitive outcomes. For study characteristics, we included the first author, publication year, sample size, average age, and sex ratio. We extracted data related to the estimation of effect size (ES) for the cognitive outcomes after exercise interventions, including the cognitive assessment tools used and the mean and standard deviation of the scores. When direct means and standard deviations were not provided in the study (for example, only the median and 95% confidence interval [CI] were provided), they were calculated according to the formula given in the Cochrane Handbook for Systematic Reviews of Interventions (Higgins, 2022).

To conduct the moderation analysis, we coded the data from each included study as follows. For the exercise moderators, we referred to the guidelines of American College of Sports Medicine (Piercy et al., 2018; Piercy and Troiano, 2018) and previous relevant reviews (Huang et al., 2022; Chen et al., 2020a; Norton et al., 2010; Chen et al., 2020b) to code the details of the exercise prescriptions into categorical variables. The exercise prescription categories included exercise type, intensity, time (duration of each bout), length of the intervention program, and frequency. To better measure exercise volume, we divided it into the duration of each bout of exercise and the length (of the overall exercise intervention program). For intensity progression, we included an intensity factor. Thus, the intensity sub-analysis comprised a progression factor in addition to three levels of intensity (low, moderate, and high). The codes and classifications for the different exercise interventions are presented in **Supplement 3**.

For the cognitive moderators, we included global cognition, executive function, memory, attention, and information processing. Global cognition is an umbrella term that refers to various areas of cognitive performance (Huang et al., 2022). Executive function involves higher cognitive processes, such as conflict management, inhibitory control, and decision-making. Memory refers to the process of acquiring and storing information in the brain. Attention is a basic cognitive ability in humans, and almost all aspects of human behavior are closely related to attention processes (Harvey, 2019). Information processing is a basic neurocognitive function that acts as a prerequisite for executive functions to emerge (Haverkamp et al., 2020). The details of the cognitive assessment tools used in the included studies are presented in **Supplement 4**.

For the age moderators, the age of the participants was coded into three categories based on the reports in each included study: children (6–17 years), younger adults (18–60 years) and older adults (>60 years).

Finally, previous research has indicated that the characteristics of the control condition(s) play a role in the magnitude of the effect of exercise interventions on cognitive functions (Northey et al., 2018). Hence, the control condition was classified into six categories based on the included studies: wait list, usual care, health education, stretching, balance and toning, and active activities.

2.4. Bias assessment

The risk of bias was assessed by two authors (MG-Z and JF-J) using the Cochrane Risk of Bias tool according to the Cochrane Handbook for Systematic Reviews of Interventions (Higgins, 2022). We selected six appraisal criteria, including random sequence generation, allocation concealment, blinding of the outcome assessor, incomplete data, selective reporting, and other sources of bias.

2.5. Statistical analysis

The statistical analysis was conducted with R v4.2.2 software using the *metafor* package 4.2.3 (Fisher and Tipton, 2010). Standard mean differences (SMDs) were calculated using the change in means and standard deviations and the number of participants after exercise interventions to estimate ES values. A positive ES value indicated that the cognitive benefit of the exercise group exceeded that of the control group; a negative ES value indicated that the cognitive improvement in the control group was greater than that in the exercise group. ESs were classified based on information in the Cochrane Handbook for Systematic Reviews of Interventions (Higgins, 2022) as small (SMD <0.40), moderate (SMD = 0.40–0.70), or large (SMD >0.70).

A multilevel random-effects model with a restricted maximum likelihood estimator was used to account for dependency between ESs (Weisz et al., 2013). This multilevel meta-analysis explains the dependency by fitting the study term as a random factor in the model, rather than requiring independency of ESs as in a traditional meta-analysis. The multilevel model was first used to assess the overall ES of exercise interventions on cognition compared with the control group. Furthermore, a mixed-effects model was fitted to determine the moderators of different exercise variables, cognitive domains, age, and control conditions as potential sources of variance. First, separate models were fitted to assess the main effects of each exercise variable, cognitive domain, age, and control condition. Specifically, how the various exercise variables would influence cognitive functions (e.g., whether aerobic exercise would have more positive effects on cognitive function compared with other exercise types or whether moderate exercise intensity would be a more suitable intensity option to improve cognitive function than other intensity options) according to FITT-VP principles. Second, after an analysis of the main effects, in order to determine whether the exercise and age moderators could influence the effects of exercise interventions on cognitive functions, the moderators that were found to be significant were added to one model to examine the potential interaction effects (Weisz et al., 2013). For these analyses, the Q statistic was calculated to assess heterogeneity, with $p < 0.05$ suggesting statistically significant heterogeneity (Higgins et al., 2003). The analysis of the main effects was interpreted using the 95% CI for the point estimates of each level of a moderator and the statistical significance of the omnibus test, with $p < 0.05$ suggesting statistical significance. Publication bias was assessed using Egger's test and visual inspection of a funnel plot (Egger et al., 1997).

3. Results

3.1. Description of included studies

A total of 18,852 studies were initially identified through database searching. The PRISMA study selection process is outlined in Fig. 1. After the duplicate and irrelevant studies (including 11 potentially eligible articles were excluded due to a lack of access to the full text) were removed based on the inclusion and exclusion criteria, 58 studies published between 2007 and 2022 were included in the qualitative analysis. Four studies (Dao et al., 2013; Liu-Ambrose et al., 2012; Tao et al., 2017; Gothe et al., 2017) were secondary analyses from one study that produced more than one publication; thus they were regarded as one study for our quantitative analysis. Hence, 54 studies with 6277 participants were included in the final quantitative meta-analysis. Among the final included studies assessing the effects of exercise interventions, 11 studies with 1751 participants analyzed global cognition (Alghadir et al., 2016; Arrieta et al., 2020; Balazova et al., 2021; Carta et al., 2021; Coelho-Junior et al., 2020; Geard et al., 2021; Kamegaya et al., 2014; Muscari et al., 2010; Santos et al., 2020; Calatayud et al., 2015; Zotcheva et al., 2021), 33 studies with 3062 participants analyzed executive function (Arrieta et al., 2020; Kamegaya et al., 2014; Antonio Parraga-Montilla et al., 2021; Brown et al., 2021; van den Berg et al., 2019; Cekanauskaite et al., 2020; Coetsee and Terblanche, 2017; Costigan et al., 2016; Eather et al., 2019; Fallah et al., 2013; Frost et al., 2021; Gothe et al., 2014; Kimura et al., 2010; Klusmann et al., 2010; Akko, 1 et al., 2020), 23 studies with 2515 participants analyzed memory (Arrieta et al., 2020; Coelho-Junior et al., 2020; Santos et al., 2020; Brown et al., 2021; Cekanauskaite et al., 2020; Gothe et al., 2014; Klusmann et al., 2010; Kamiyo et al., 2011; Kosmat and Vranic, 2017; Leahy et al., 2020; Liu-Ambrose et al., 2010; Ludyga et al., 2018; Meijer et al., 2020; Moreau et al., 2017; Mortimer et al., 2012; Nouchi et al., 2014; Ruscheweyh et al., 2011; Erickson et al., 2011; Sexton et al., 2020; Taylor-Piliae et al., 2010; Tao et al., 2016; Cassilhas et al., 2007; Latino et al., 2021), 9 studies 1659 participants analyzed attention (Antonio Parraga-Montilla et al., 2021; Brown et al., 2021; van den Berg et al., 2019; Moreau et al., 2017; Latino et al., 2021; Li et al., 2015; Qi et al., 2021; Linde and Alfermann, 2014; Solianik et al., 2021), and 7 studies with 451 participants analyzed information processing (Arrieta et al., 2020; Cekanauskaite et al., 2020; Gothe et al., 2014; Ruscheweyh et al., 2011; Taylor-Piliae et al., 2010; Linde and Alfermann, 2014; Wang et al., 2019). Study characteristics, exercise prescriptions, and cognitive tests of interest are given in Supplement 5.

3.2. Risk of bias assessment

We evaluated the risk of bias in the included studies. The percentages of bias risk (high, unclear, or low) are described here and shown in Fig. 2: random sequence generation (100% low risk), allocation concealment (0 for high risk, 48.2% for unclear, 51.8% for low risk), blinding of outcome assessment (3.7% for high risk, 83.3% for unclear, 13.0% for low risk), incomplete outcome (20.3% for high risk, 9.3% for unclear, 70.4% for low risk), selective outcome reporting (100% for low risk), and other risks of bias (3.7% for high risk, 5.6% for unclear, 90.7% for low risk).

3.3. Overall Analyses and Publication Bias

The results of our overall analysis of the effects of exercise interventions on cognitive function indicated a significant positive ES (SMD=0.39, 95% CI: 0.24–0.54), and there was significant heterogeneity present ($Q_{82}=620.97$, $p < 0.001$). The forest plot of the effects of exercise interventions on overall cognition and its domains is presented in Fig. 3.

Note: GC=global cognition, EF=executive function, MF=memory, AT=attention, IP=information processing.

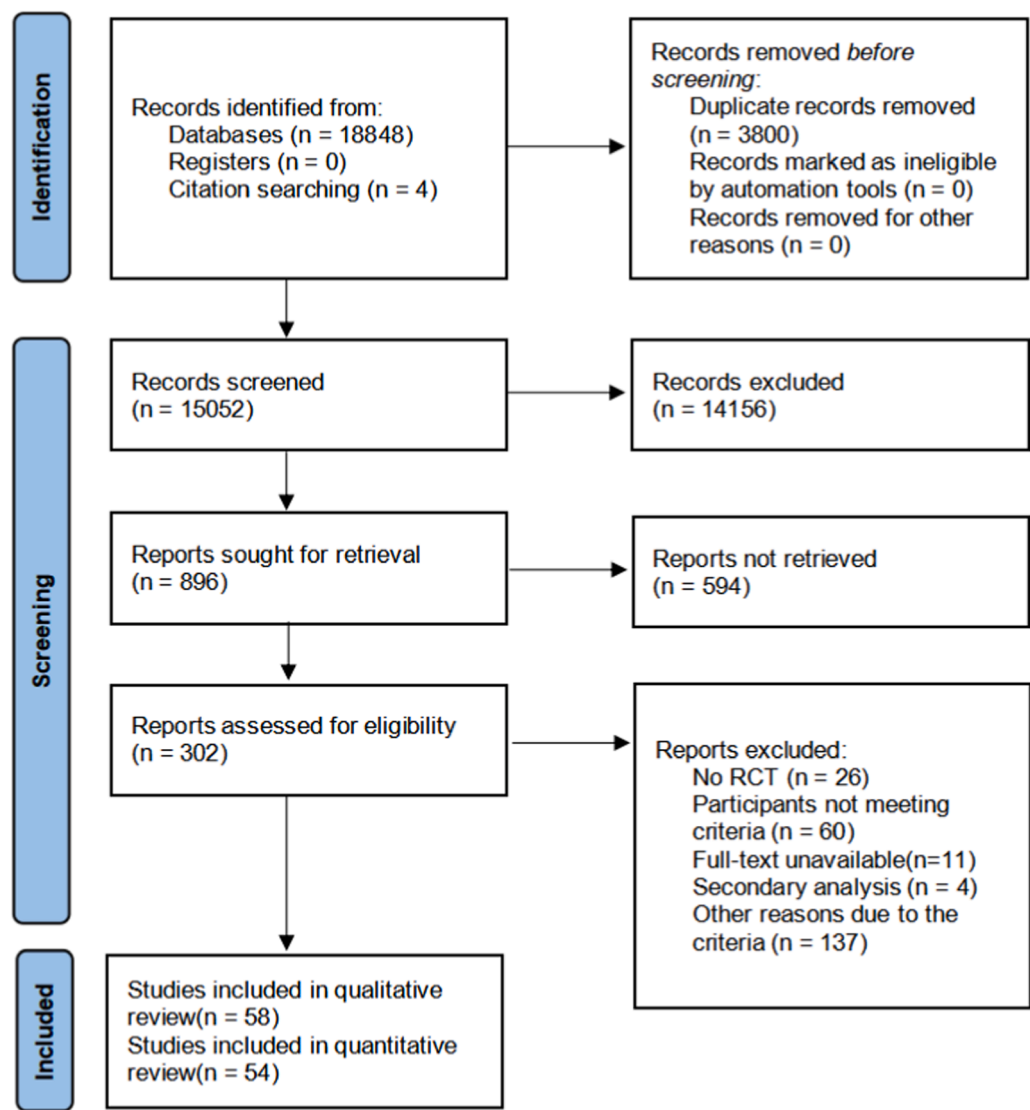


Fig. 1. Study selection.

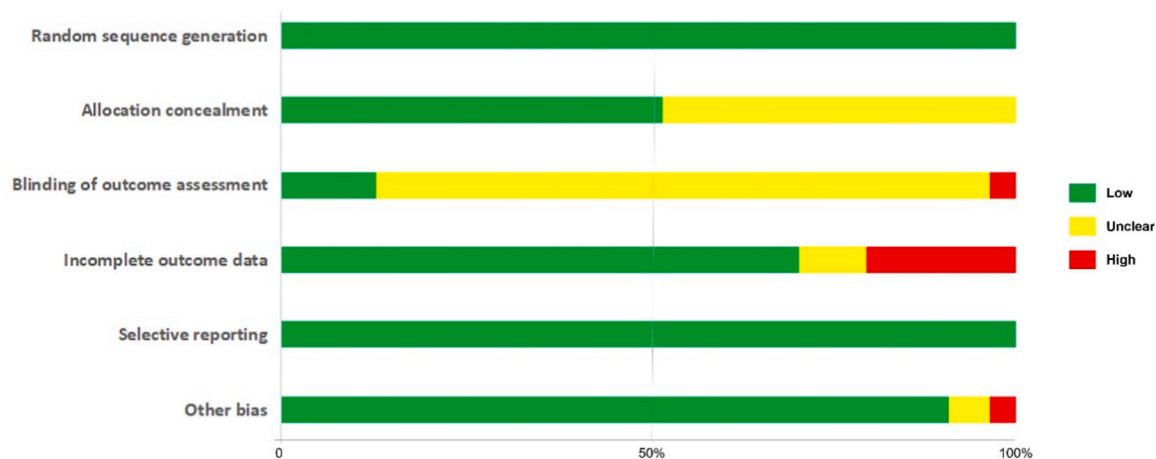


Fig. 2. Risk of bias indicated by percentages of assessed biases across all included studies.

The results of Egger’s test suggested that there was no relationship between the standard error and ES ($t = 1.74, p = 0.06$). In addition, a visual inspection of the funnel plot indicated that the included samples

did not systematically affect the findings. The funnel plot is presented in **Supplement 6**.

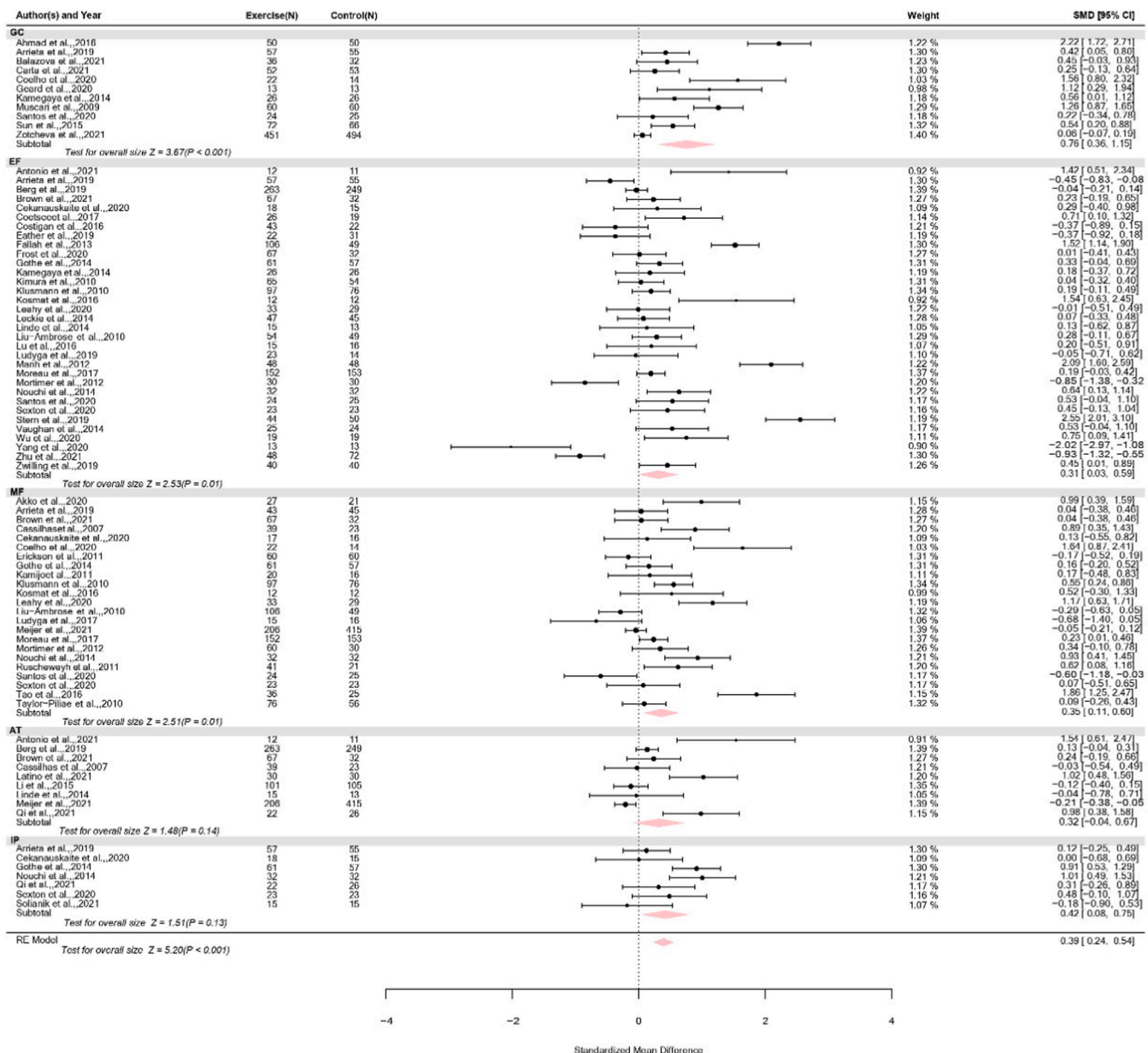


Fig. 3. Forest plot of the effects of exercise interventions on overall cognition and different cognitive domains.

3.4. Moderation analysis

To investigate the potential sources of variance, we conducted moderation analyses of exercise, age, cognition, and control conditions one at a time in separate models. The summarized analysis results are presented in Table 1, and all the forest plots of moderation analysis are presented in Supplement 7.

3.4.1. Exercise moderators

Based on the FITT-VP principles and the specific coding methods used in present study, we aimed to identify those exercise variables that moderated cognitive functions. The analysis of exercise type indicated that all exercise types had significant effects on cognition, with positive ESs and all p values < 0.01 . Resistance exercise produced the largest ES (SMD=0.51, 95% CI: 0.12–0.90). For exercise duration, moderate and long exercise durations ($p < 0.001$) both had significant effects on cognition, with positive ESs, but moderate exercise duration produced a larger ES (SMD=0.42, 95% CI: 0.23–0.64). The analysis of exercise

frequency indicated that low and moderate exercise frequency both had significant effects on cognition with positive ESs, but a low frequency of exercise produced a larger ES and lower p values (SMD=0.43, 95% CI: 0.19–0.68, $p < 0.001$). For exercise intensity, all exercise intensities produced significant and positive ESs, except for intensity progression, but moderate exercise intensity had the largest ES and strongest significance (SMD=0.56, 95% CI: 0.27–0.85, $p < 0.001$) compared with low and high exercise intensity. Finally, for the overall length of the exercise, long length interventions produced positive but insignificant ES, short and moderate exercise lengths both had significant positive ESs, but moderate length (SMD=0.55, 95% CI: 0.30–0.79, $p < 0.001$) produced a larger ES than short exercise lengths.

3.4.2. Cognitive moderators

The results of a main effects analysis showed that some of the assessed cognitive domains benefited positively from exercise interventions. Specifically, global cognition ($p < 0.001$), executive function ($p = 0.01$), and memory ($p = 0.01$) showed statistically significant

Table 1
Summary of the results of the moderator analysis.

Moderator	Effect size (n)	Estimate Mean (95% CI)	Q statistic	Omnibus test of moderators
Exercise moderators				
Type	$Q_{86} = 598.05, P < 0.001$	$Q_4 = 31.50, P < 0.001$		
AE	26	0.44 (0.18, 0.70)		
RE	12	0.51 (0.12, 0.90)		
ME	24	0.29 (0.02, 0.57)		
MBE	28	0.41 (0.15, 0.67)		
Duration	$Q_{80} = 586.59, P < 0.001$	$Q_3 = 20.04, P < 0.001$		
Short	23	0.19 (-0.11, 0.49)		
Moderate	42	0.42 (0.23, 0.64)		
Long	18	0.35 (0.02, 0.69)		
Frequency	$Q_{79} = 593.87, P < 0.001$	$Q_3 = 25.31, P < 0.001$		
Low	32	0.43 (0.19, 0.68)		
moderate	39	0.35 (0.13, 0.57)		
High	11	0.37 (-0.03, 0.79)		
Intensity	$Q_{80} = 621.35, P < 0.001$	$Q_4 = 28.02, P < 0.001$		
Low	21	0.40 (0.09, 0.71)		
Moderate	23	0.56 (0.27, 0.85)		
High	24	0.40 (0.11, 0.68)		
Progression	16	0.12 (-0.22, 0.47)		
Length	$Q_{79} = 599.04, P < 0.001$	$Q_3 = 29.28, P < 0.001$		
Short	41	0.32 (0.11, 0.54)		
Moderate	31	0.55 (0.30, 0.79)		
Long	10	0.24 (-0.17, 0.66)		
Cognitive moderators				
Overall	83	0.39 (0.24, 0.54)	$Q_{82} = 620.97, P < 0.001$	
Cognitive domains	$Q_{78} = 594.41, P < 0.001$	$Q_5 = 30.62, P < 0.001$		
GC	11	0.76 (0.35, 1.15)		
EF	33	0.31 (0.07, 0.59)		
MF	23	0.35 (0.11, 0.60)		
AT	9	0.32 (-0.04, 0.67)		
IP	7	0.42 (0.08, 0.75)		
Age (years)	$Q_{80} = 568.46, P < 0.001$	$Q_3 = 30.56, P < 0.001$		
6–17	14	0.18 (-0.17, 0.53)		
18–60	6	0.40 (-0.15, 0.94)		
> 60	63	0.45 (0.28, 0.62)		
Control conditions	$Q_{77} = 598.92, P < 0.001$	$Q_6 = 26.75, P < 0.001$		
Usual care	32	0.39 (0.14, 0.65)		
Wait list	17	0.26 (-0.07, 0.61)		
Health education	14	0.38 (0.21, 0.75)		
Active	9	0.42 (-0.03, 0.87)		
Balance and toning	3	0.50 (-0.27, 1.27)		
Stretching	8	0.57 (0.09, 1.05)		

Abbreviations: AE, aerobic exercise; MBE, mind-body exercise; ME, multicomponent exercise; RE, resistance exercise. GC, global cognition; EF, executive function; MF, memory; AT, attention; IP, information processing.

differences compared to the control groups, whereas no statistical significance was found for attention ($p = 0.14$) and information processing ($p = 0.13$).

This study aimed to explore the roles that different exercise variables (including type, frequency, intensity, duration, and length, according to FITT-VP principles) play in the magnitude of the effects of exercise interventions on cognitive domains. Previous related reviews have indicated that the effects of exercise interventions on cognition may vary depending on the exercise variable and cognitive domain assessed. Therefore, we also included the moderators of exercise and cognition as interactive terms in separate models.

For global cognition, aerobic exercise (SMD=1.13, 95% CI: 0.43–1.83; $p < 0.01$), moderate duration (SMD=0.92, 95% CI: 0.41–1.42; $p < 0.001$), moderate frequency (SMD=0.88, 95% CI: 0.27–1.49; $p < 0.01$), moderate intensity (SMD=1.04, 95% CI: 0.39–1.69; $p < 0.01$), and moderate length of intervention (SMD=1.01, 95% CI: 0.39–1.62; $p < 0.01$) all had significant interaction effects on global cognition.

For executive function, resistance exercise (SMD=0.61, 95% CI: -0.01 to 1.24; $p < 0.05$), low frequency (SMD=0.43, 95% CI: 0.03–0.83; $p < 0.01$), and moderate length intervention (SMD=0.55, 95% CI: 0.15–0.95; $p < 0.01$) had significant interaction effects on executive function. A long duration of a single bout of exercise produced the largest effect but was only marginally significant (SMD=0.46, 95% CI: -0.05 to 0.97; $p = 0.07$). Similar moderate ESs were found for both moderate intensity exercise (SMD=0.42, 95% CI: -0.09 to 0.94; $p = 0.11$) and exercise intensity progression (SMD=0.42, 95% CI: -0.22 to 1.06; $p = 0.20$) although neither were statistically significant.

In terms of memory, mind-body exercise (SMD=0.62, 95% CI: 0.11–1.13; $p < 0.01$), moderate duration (SMD=0.44, 95% CI: 0.01–0.86; $p < 0.05$), moderate frequency (SMD=0.40, 95% CI: 0.01–0.79; $p < 0.05$), high-intensity exercise (SMD=0.66, 95% CI: 0.11–1.22; $p < 0.05$) and moderate intervention length (SMD=0.52, 95% CI: 0.06–0.97; $p < 0.05$) all showed significant interaction effects on memory.

For attention and information processing, only low-intensity (SMD=0.79, 95% CI: 0.04–1.53; $p < 0.05$) and moderate frequency (SMD=0.81, 95% CI: 0.01–1.60; $p < 0.05$) exercise had significant interaction effects with attention and information processing, respectively. No other interaction effects were found.

3.4.3. Age moderators

Our main effects analysis of age indicated that age significantly moderated the effects of exercise interventions on cognitive function ($p < 0.001$), with only older participants significantly benefitting from exercise interventions compared with controls (SMD=0.45, 95% CI: 0.28–0.62; $p < 0.001$). To test the differential effects of exercise interventions on cognitive domains dependent on age, cognitive moderators were analyzed for assessment of an interaction effect. The results indicated that global cognition (SMD=0.73, 95% CI: 0.30–1.16; $p < 0.001$), executive function (SMD=0.40, 95% CI: 0.11–0.68; $p < 0.01$), and memory (SMD=0.38, 95% CI: 0.04–0.71; $p < 0.05$) performances were significantly improved in older participants.

3.4.4. Control condition moderators

The type of control condition was associated with differences in statistical significance. When passive control conditions were assessed (usual care, $p < 0.01$; wait list, $p < 0.05$; and health education, $p < 0.05$), the ESs were positive and statistically significant. When active control conditions were assessed (stretching, balance and toning, and active activities; all $p > 0.05$), the ESs were also positive but were not statistically significant.

4. Discussion

The present meta-analysis offers a comprehensive assessment of the effects of exercise interventions on several cognitive functions, as reported in 54 RCTs with 6277 participants. Our meta-analysis was conducted from the perspective of the guiding principles of FITT-VP exercise prescriptions (categorized herein as exercise type, duration, frequency, intensity, and length) as well as by participant age to examine their moderating effects on cognitive functions, including global cognition, executive function, memory, attention, and information processing. In general, our results demonstrated that the assessed exercise interventions had positive and statistically significant improvements on cognitive function overall and on all included cognitive domains with different ESs. The cognitive domains of global cognition benefited from exercise interventions with largely significant and positive effects, executive function and memory benefited from exercise interventions with slightly significant and positive effects, and attention and information processing showed positive benefits but without statistical significance. Compared with the two younger populations (comprising healthy individuals aged 6–60 years), the oldest population (healthy adults >60 years) showed the largest significant cognitive benefits from exercise interventions, with global cognition, executive function, and memory significantly improved compared with controls.

The main strength of our study was that it followed the FITT-VP principles, which offer scientific guidance to differentiate exercise variables. To our knowledge, this is the first study to conduct such a comprehensive meta-analysis assessing the effects of specific exercise intervention components on cognition function. Our results may be used as a reference for designing exercise prescriptions aimed at improving specific cognitive functions or preventing cognitive decline in healthy persons as well as for suggesting the direction of future exercise-cognition related studies.

4.1. Global cognition

Using an exercise coding method consistent with FITT-VP principles, we found that exercise variables as moderators played an important role to impact the effects of exercise interventions on global cognition benefits.

Different exercise types produced variously significant differences in global cognition. Our results demonstrated that all exercise types assessed were associated with improved global cognition but that aerobic exercise had a more significant interaction effect with global cognition than the other types of exercise. Previous studies have reported that resistance exercise improves global cognition in older populations with dementia or mild cognitive impairment (Gallardo-Gómez et al., 2022; Gomes-Osman et al., 2018). Our results suggested that aerobic exercise produced the largest benefits for healthy populations. Considering that few previous studies have examined the effects of exercise interventions on global cognition in healthy populations, our results offer an important opportunity for understanding this relationship. Resistance exercise also showed large effects on global cognition, but it was second to aerobic exercise in our study. Thus, our results agree with the conclusions of those studies reporting that improvement in global cognition after physical exercise intervention is the most consistent change observed among cognitive parameters for community-dwelling older adults or patients with cognitive dysfunction, (Huang et al., 2022; Gallardo-Gómez et al., 2022; Wu et al., 2019) and provides evidence in support of this effect in healthy populations. Mind-body exercise showed a moderately positive effect on global cognition, a result that was consistent with previous studies indicating that mind-body exercise presented effective improvements on global cognition among older adults with or without cognitive impairment (Bossers et al., 2015). Our results also demonstrated that multicomponent exercise produced moderate effects on global cognition, supporting previous meta-analyses showing that combined exercise programs have moderate effects on

global cognitive function (Northey et al., 2018; Wu et al., 2019). Multicomponent exercise was defined as consisting of two or more types of exercise, including aerobic and resistance exercise as well as others. We believe that multicomponent exercise weakens the effect of the exercise intervention compared with resistance exercise alone. For example, a previous study pointed out that for a 30-minute aerobic exercise and a 30-minute multicomponent exercise (including the same aerobic exercise but adding resistance exercise) lasting 4 weeks, the duration and frequency of the aerobic exercise and of the resistance exercise are each reduced by half in the multicomponent exercise compared with only a 30-minute aerobic exercise lasting 4 weeks, thus weakening the cognitive benefits (Colcombe and Kramer, 2003).

In addition to exercise type, other exercise factors included in the FITT-VP principles also produced significant interaction effects on global cognition. Our findings from moderation analyses indicated that global cognition benefits were greatest for moderate intensity exercise performed 3–4 times per week, with bout durations of 45–60 min for a total of 3–6 months. These findings are consistent in part with those recommended by the ACSM and with previous meta-analyses. For example, Northey, Cherbuin, Pampa, Smeed and Rattray (Northey et al., 2018) suggested that at least moderate intensity and at least 45 min per session on as many days of the week as possible is beneficial to global cognitive function in adults aged > 50 years. Moderate exercise bout durations also yielded positive cognitive improvements in another study (Xue et al., 2019). In the present meta-analysis, except for one study with adult participants 40–50 years of age, most of the participants were older adults (mean age >60 years). Thus, the benefits of exercise interventions on global cognition were primarily assessed in older adults.

Therefore, based on our moderation analysis results of FITT-VP categories, we suggest that aerobic exercise with a moderate frequency, duration, overall length, and intensity show the most promise for improving global cognition in older adults.

4.2. Executive function

The results of the current meta-analysis indicated that exercise interventions may improve executive function but with small ESs. These results are consistent with a handful of previous meta-analyses that also indicated that exercise interventions promoting executive function had small ESs in adolescent, young adult, and older adult populations (Chen et al., 2020a; Haverkamp et al., 2020; Northey et al., 2018; Falck et al., 2019; Di Lorito et al., 2021). Our moderation analysis suggested that the diverse exercise variables included in the FITT-VP principles play important roles in moderating intervention effects on executive function.

In terms of exercise type, our results indicated that both resistance and aerobic exercise were significantly associated with enhanced executive function. Consistent with this finding, two recent meta-analyses also showed that resistance exercise appeared to be the most effective way to improve executive function (Huang et al., 2022; Yu et al., 2018), and two other studies found that aerobic exercise played an important role in improving executive function (Xue et al., 2019; Wang et al., 2021). Interestingly, it seems that our results may support these two different exercise types. However, we found a slight difference between aerobic and resistance exercise, with resistance exercise showing larger positive effect sizes than aerobic exercise. Thus, we suggest that resistance exercise is a more effective exercise type than aerobic exercise for improving executive function. We found that mind-body exercise had a moderately positive effect on executive function. A handful of previous mind-body exercise studies offered positive evidence to recommend this exercise as improving executive function or its sub-domains performances of various populations (Zhang et al., 2018; Lin et al., 2022; Ren et al., 2021; Chan et al., 2019). The mind-body exercises assessed in the studies included in the present meta-analysis consisted of yoga and tai chi among others that have been shown to influence various aspects of physical fitness, such as flexibility or balance (Kaushal et al., 2018).

These types of exercise emphasize the connection between physical activity and the mind to strengthen self-control abilities, which may play a positive role in executive functions. For multicomponent exercise, a small and positive effect showed in our results. In addition to the different mechanisms of intervention when combining various exercise components, participants need to coordinate and control different exercise modes during exercise, which may also have positive effects on cognitive function, such as executive conversion and information updating (Sanders and Hortobágyi, 2019). We encourage new RCTs and meta-analysis studies to further explore this issue, especially, to differentiate the most effective types of aerobic and resistance exercises.

Previous meta-analyses have rarely explored exercise dose-response relationships of executive function among children, adolescent, and adult populations consistent with exercise prescription principles. Thus, we conduct a moderation analysis to fill this gap based on FITT-VP principles. Regarding the dose-response relationship of other exercise characteristics, our results showed that low exercise frequency (1–2 times/week) for 3–6 months of moderate intervention length were associated with the largest executive function benefits for healthy participants. These findings are partly consistent with a previous meta-analysis that showed that positive executive function improvements were observed within these ranges (Kelly et al., 2014; Wu et al., 2019; Falck et al., 2019), especially in older adults. For exercise duration, although the interaction of moderate duration and executive function presented with a marginally significant effect ($p = 0.07$), given the adequate ESs of moderate duration compared with long and short durations, we still recommend it as an effective exercise duration option to improve executive function although additional studies will be needed to strengthen this finding. For exercise intensity, we found that both intensity progression and moderate intensity had similar executive function benefits, based on their ESs. During the exercise intervention sessions, exerting different exercise intensities at different stages is more of a challenge for physical activity levels and cognition compared with a single intensity. However, moderate intensity has also been shown to be effective for improving executive function in most previous studies (Chen et al., 2020a; Falck et al., 2019). Given that intensity progression was not included as a moderating factor analyzed in many previous studies, there was insufficient evidence to determine whether it or moderate intensity would be the better intensity option. A previous review attempted to assess the benefits of intensity progression on brain function, but found it difficult owing to the small number of studies (Chen et al., 2020b). Therefore, more studies will be needed exploring the effects of exercise intensity progression and moderate intensity on executive function.

4.3. Memory

Although previous studies have found that memory is the least improved cognitive domain in response to exercise interventions (Huang et al., 2022; Rathore and Lom, 2017), our results indicated a moderate and significant effect. Furthermore, moderations analysis of interaction effects suggested that some exercise prescriptive options based on FITT-VP guidelines offer greater memory benefits.

For exercise type, our results showed that mind-body exercise had the largest positive benefit on memory in our healthy population, followed by resistance exercise. Previous studies have suggested that of the types of exercise assessed, only resistance exercise offered significant benefits for memory among patients with mild cognitive impairment. Although resistance exercise also had positive benefits, second only to mind-body exercise on memory in our results, mind-body exercise still showed effects superior to resistance exercise, and resistance exercise did not show a statistically significant difference compared with controls. We speculate that this difference between study results may be due to the exercise and sample population characteristics. First, mind-body exercise, such as tai chi or baduanjin, typically requires participants to remember a series of complex body movements and to move the trunk

and all four limbs (Tao et al., 2017). The challenge of remembering the patterns and sequences of these movements offers an opportunity to enhance memory. Second, our study included only healthy participants with normal cognitive function, rather than participants with mild cognitive impairment, resulting in differing initial memory abilities. In addition, only three studies that assessed the effects of resistance exercise on memory were included in our meta-analysis, which may have contributed to our finding of a weaker association. Hence, more studies with resistance exercises and mind-body exercises are needed to explore the cognitive benefits and differences between populations with different cognitive statuses.

Regarding the interaction effects of the other exercise characteristics of FITT-VP guidelines on memory, our moderation analysis results indicated that a moderate exercise frequency of 45–60 min per week for a total intervention of 3–6 months may benefit memory. These results were consistent in part with a previous study (Northey et al., 2018). We also found that higher exercise intensity offered greater memory benefits. It is unclear how exercise intensity affects memory or working memory. One previous meta-analysis indicated that chronic exercise with various intensity loads affects working memory, whereas other studies indicated that the beneficial effects of physical activity on memory function were independent of intensity (Meijer et al., 2020; Falck et al., 2019; Álvarez-Bueno et al., 2017; Esteban-Cornejo et al., 2015). Given the incomplete evidence, it is important for more studies to be conducted to test the dose-response effects of these exercise characteristics on memory benefits.

4.4. Attention and information processing

The present meta-analysis found positive but not significant overall ESs for exercise interventions both on the cognitive domains of attention and information processing. However, caution is warranted in interpreting the findings of our subgroup analyses of the exercise prescriptions with these cognitive domains because of the small study samples. One previous meta-analysis did not present the results of subgroup analyses when fewer than 10 studies were assessed (Chen et al., 2020a). We chose to present the results of the moderation analyses despite the small number of studies to offer a better understanding of how exercise interventions may affect attention and information processing. However, the small sample sizes may be insufficient to draw conclusions for all exercise characteristics. For example, only one study examined the effects of moderate exercise frequency or moderate-intensity exercise. For information processing, only studies assessing the effects of aerobic, mind-body exercise and multicomponent exercise were included in our analyses, and most of the data on the exercise characteristics examined were from one study each. The sample sizes of previous meta-analyses may have been larger because they included studies that (1) combined exercise with cognitive training or video games, (2) were observational or cluster RCTs, rather than only RCTs, and (3) analyzed acute exercise interventions (Kelly et al., 2014; Chang et al., 2012; Haverkamp et al., 2020; Yu et al., 2018; Walsh and Tschakovsky, 2018). Our goal was to assess the cognitive benefits of chronic exercise interventions without interference from other interventions (such as videogame or cognitive training) and based on FITT-VP principles.

In summary, the present meta-analysis results suggested that exercise interventions had overall positive effects on attention and information processing. However, additional well-designed RCT studies guided by FITT-VP principles are needed.

4.5. Age and control conditions

We also aimed to investigate whether age played a role in the effects of exercise interventions on cognition. We included studies with participants aged from 6 years to > 60 years and grouped them by their average ages as follows: 6–17 years (children and adolescents), 18–60

years (adults), and > 60 years (older adults). However, during article screening and analysis, we found that most of the articles included predominately older adult populations and assessed the cognitive domains of executive function or memory function. Fewer studies assessed younger populations or attention or information processing. For example, of 54 articles included in this meta-analysis, 38 examined older adults, whereas only 10 investigated children and adolescents, and 6 assessed adults. Among the cognitive domains, executive function accounted for the largest research interest, global cognition was examined mainly in older adults, and attention and information processing were studied the least. Given our strict inclusion criteria, we expected to exclude several articles during the screening process that combined exercise interventions with videogames or included non-RCTs, but we discovered that most studies also examined acute exercise rather than chronic exercise for adult populations. The results of our funnel plot showed that there was no relationship between the standard error and ES (SMD), suggesting that the small sample sizes did not systematically affect our findings (Egger et al., 1997). Still, we recommend additional RCTs be conducted that take these issues into consideration.

In the present study, according to our moderation analysis of age and cognitive domains, exercise interventions played a positive role in improving global cognition, executive function, and memory especially among older participants (>60 years), offering firm support for cognitive benefits from exercise interventions in older populations. This finding encourages research aimed at advancing the understanding of age-related cognitive benefits attributable to exercise interventions. That exercise interventions were associated with cognitive benefits in older participants suggests that more attention should be paid to exercise interventions in adult participants to maximize the effects of exercise interventions in promoting cognitive improvement and providing a more adequate cognitive capacity reserve for individuals during age-related cognitive decline. We suggest that future studies explore this issue.

Finally, a previous study indicated that the overall effects of exercise interventions on cognitive function were present only when the comparison was with passive controls (such as wait list or health education), with no statistically significant effects of exercise interventions when compared with active controls (such as stretching) (Northey et al., 2018). The results of our moderation analysis of control conditions supported the findings of that study. Hence, the moderating effects suggest that control conditions are an important factor influencing the cognitive benefits of exercise interventions and that more RCT studies are needed to go beyond potential confounding factors, such as socialization and attention, in order to identify the optimal type of exercise prescription for maximizing cognitive benefits.

4.6. Underlying mechanisms

Long-term regular exercise has a positive effect on stimulating brain plasticity and improving cognition and brain function (Wrann et al., 2013; Ashpole et al., 2015). A range of molecular mediators, such as brain-derived neurotrophic factor (BDNF), insulin-like growth factor 1 (IGF-1), and irisin, are potential mechanisms whereby exercise induces those cognition and brain benefits. Although these mediators play potential bio-mediation roles in the relationship between exercise interventions and cognitive function, their roles may differ across the lifespan (Jandova et al., 2021; Siracusa et al., 2020). Hence, given the fluctuation of these biomarkers across the lifespan, exercise interventions and the potential bio-mechanisms may play different roles to produce effects. In summary, we emphasize once again that offering more exercise interventions to less functional older populations may induce more biological mediators to help guard against cognitive decline during aging. Additionally, there is increasing evidence that oxidative stress contributes to the pathogenesis and progression of neurodegenerative diseases, such as Parkinson's disease and vascular dementia (Calabrese et al., 2016; Mancuso et al., 2008). Modulating endogenous cellular

defense mechanisms through stress response signaling is an innovative approach to treating chronic neurodegenerative diseases or cancer. This biologically relevant system offers new insights on age-related disease and on the longevity process (Calabrese et al., 2007; Mancuso et al., 2008). We encourage more relevant studies to explore these issues.

4.7. Implications and limitations

The present systematic review and meta-analysis offers some important considerations for future studies. First, aerobic and resistance exercise should be included in RCTs assessing the effects of exercise on global cognition and executive function. Second, RCTs assessing cognitive benefits based on exercise prescriptions should include younger adults, rather than only older adults, and healthy populations, rather than only individuals with deficits in a specific cognitive domain. Furthermore, future studies should be designed to compare head-to-head the effects of different factors of exercise interventions on cognition, for example, an RCT to compare resistance exercise against aerobic exercise for executive function according to our results. Lastly, the cognitive benefits of exercise interventions should be compared between different age groups to provide more evidence for understanding exercise programming for cognitive decline in aging.

This meta-analysis has some limitations. First, many studies were excluded owing to our strict inclusion criteria; thus, those data were not taken into account in our interpretations of our data. Second, we extracted only cognitive behavioral data and did not assess underlying mechanisms or brain activity. Third, we collected only articles published in English, which may have led to missing some information. Additionally, some studies were not included into further analysis because of the full-text unavailable, this methodological limitation may also have resulted in potentially eligible studies being missed. Fourth, only two databases were searched in our study, which may lead to some relevant studies not being included. Finally, substantial heterogeneity existed across our results, which may weaken the certainty of our evidence, and this heterogeneity should be considered when interpreting our results. The potential reasons for the high heterogeneity may be the different populations, control conditions, and cognitive assessment tools used in each included study. We suggest that these factors be considered to better control heterogeneity in future studies.

5. Conclusion

This systematic review and meta-analysis extracted cognitive outcomes from previous RCTs that assessed the effects of chronic exercise interventions on various cognitive domains in healthy populations from 6 to > 60 years of age and analyzed the exercise variables based on FITT-VP principles. Our results from moderation analyses demonstrated that aerobic exercise performed with moderate duration, frequency, intensity, and overall length was associated with the greatest improvement in global cognition. Resistance exercise conducted with a long session bout at a low weekly frequency and moderate length of time overall also improved executive function, but future studies will be needed to clarify the exercise effect differences between the use of a moderate exercise intensity and intensity progression. Mind-body exercise with moderate duration, frequency, and overall length but high intensity was associated with improvements in memory. Our results assessing attention and information processing should be interpreted cautiously because of the low number of included studies. Finally, older participants benefited most from exercise interventions. This study offers new insights on the dose-response relationship of chronic exercise and the use of FITT-VP exercise principles to improve cognitive abilities or prevent cognitive decline in the process of aging.

Author contributions

MG-Z and XC-W were responsible for the conception, design and

hypothesis. MG-Z, JF-J, Y-Y and LP-Z conducted the systematic search, data extraction, and the assessment of risk of bias. MG-Z and JF-J completed all statistical analyses. MG-Z and XC-W contributed to the drafting of the manuscript and its revisions, including the final version. All authors read and approved a draft and the final manuscript.

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Declaration of Competing Interest

None.

Data Availability

Data will be made available on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.arr.2023.102116](https://doi.org/10.1016/j.arr.2023.102116).

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